

A simple primitive axial algebra with a disconnected non-annihilation graph, and a decomposable axial block

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Abstract

An axial algebra is a commutative non-associative algebra generated by idempotents, called axes, whose adjoint actions are semisimple, with eigenvectors that multiply according to a fusion law. McInroy and Shpectorov conjectured that the finest sum decomposition of an axial algebra comes from the connected components of its non-annihilation graph $\Delta(X)$, whose vertices are the generating axes, two of them being adjacent when their product is non-zero; Gorshkov and Shpectorov, and Mamontov, Shpectorov and Zhelyabin, ask the same question for primitive axial algebras. We show that, when arbitrary finite symmetric fusion laws are allowed, the answer is negative, even for simple algebras. A four-dimensional simple algebra S over $\mathbb{Q}$ is generated by three primitive axes a, b, c for the law $\mathcal{J}^+(\frac{1}{2})$, which is the Jordan-type law $\mathcal{J}(\frac{1}{2})$ with $0 \star 0$ and $0 \star \frac{1}{2}$ enlarged to $\{1, 0, \frac{1}{2}\}$; its graph $\Delta(X)$ has the components $\{a, b\}$ and $\{c\}$, the subalgebras they generate do not annihilate each other, and S has no non-trivial sum decomposition. We also give a five-dimensional primitive axial algebra, for a similar law $\mathcal{J}^\circ(\frac{1}{2})$, in which the block of an axis (the ideal it generates) is the sum of two of its proper ideals; when arbitrary finite symmetric fusion laws are allowed, this answers a question of Mamontov, Shpectorov and Zhelyabin negatively. The laws $\mathcal{J}^+(\frac{1}{2})$ and $\mathcal{J}^\circ(\frac{1}{2})$ are neither Seress nor of Jordan type, so the known positive results do not apply to them, and we claim nothing about algebras of Monster type, Seress or graded fusion laws, or Majorana algebras. We also point out that a known two-dimensional algebra of Peng and of Kaygorodov, Martín González and Páez-Guillán has properly nested blocks, which answers a question of Mamontov, Shpectorov and Zhelyabin on dominance; a further four-dimensional example has nested blocks and a disconnected graph. These results are formally verified in Lean 4 with Mathlib. The work was done by AI agents under the author's direction; their role is described at the end of the paper.

1 Introduction

Background. Axial algebras were introduced by Hall, Rehren and Shpectorov [3, 4]. An axial algebra is a commutative, not necessarily associative algebra generated by idempotents, called axes, whose adjoint actions are semisimple and whose eigenvectors multiply according to a prescribed fusion law; the definitions are recalled in Section 2. The surveys of McInroy and Shpectorov [10] and of Gorshkov and Shpectorov [2] describe the theory and collect open problems.

Let (A, X) be a primitive axial algebra, X being a set of primitive axes that generates A . The *non-annihilation graph* $\Delta(X)$ has vertex set X , distinct axes a, b being adjacent when $ab \neq 0$ [7, Def. 6.1], [10, Def. 3.15]. A *sum decomposition* of A is a set of pairwise annihilating subalgebras that generates A [7, Def. 5.1], [10, Def. 3.11]. Khasraw, McInroy and Shpectorov conjectured that for axial algebras of Monster type $\mathcal{M}(\frac{1}{4}, \frac{1}{32})$ the finest sum decomposition arises

when each part corresponds to a single connected component of $\Delta(X)$ [7, Conj. 6.2]. McInroy and Shpectorov stated this conjecture for axial algebras in general [10, Conj. 3.16] and observed that, if X_i are the connected components and $B_i = \langle\langle X_i \rangle\rangle$ are the subalgebras they generate, it amounts to $B_i B_j = 0$ for $i \neq j$. Gorshkov and Shpectorov [2, Problem 3.11] and Mamontov, Shpectorov and Zhelyabin [8, Question 9.5] ask whether, in the finest sum decomposition of a primitive axial algebra, the summands correspond to the connected components of Δ ; the authors of [8] restate this as the question whether an indecomposable primitive axial algebra can have a disconnected graph Δ , and single out simple Majorana algebras [8, Question 9.6], [2, Problem 3.12].

The conjecture holds for algebras of Jordan type η by [5, Theorem A], as recorded in [7, 10]. For Seress fusion laws (Definition 2.1) it holds when all but at most one of the subalgebras B_i are slender [7, Thm 6.14], [10, Thm 3.20], and the authors of [10] remark that few non-slender axial algebras are known. For Majorana algebras, indecomposable algebras are simple [8].

Results. The questions of [2, 8] are posed for primitive axial algebras with an arbitrary fusion law; in [7] fusion laws are finite and symmetric, and from Section 2.1 on [10] considers only symmetric ones. We show that the answer is negative as soon as arbitrary finite symmetric fusion laws are allowed. For $\eta \neq 0, 1$ let $\mathcal{J}^+(\eta)$ be the fusion law on $\{1, 0, \eta\}$ that agrees with the Jordan-type law $\mathcal{J}(\eta)$ except that $0 \star 0$ and $0 \star \eta$ are enlarged to $\{1, 0, \eta\}$ (Table 1).

Theorem A. *Let S be the four-dimensional commutative algebra over $\mathbb{Q}$ with basis a, b, x, c and products*

$$\begin{aligned} a^2 &= a, & b^2 &= b, & x^2 &= x, & c^2 &= c, \\ ab &= \frac{1}{4}(a + b - x), & ax &= \frac{1}{4}(a - b + x), & bx &= \frac{1}{4}(-a + b + x), \\ ac &= bc = 0, & cx &= \frac{1}{2}(-a - b + x) - \frac{1}{4}c, \end{aligned}$$

and let $X = \{a, b, c\}$. Then:

- (a) (S, X) is a primitive $\mathcal{J}^+(\frac{1}{2})$ -axial algebra;
- (b) S is simple;
- (c) $\Delta(X)$ is not connected: its only edge is $a-b$;
- (d) every sum decomposition of S is trivial, that is, contains S ;
- (e) the subalgebras $\langle\langle a, b \rangle\rangle$ and $\langle\langle c \rangle\rangle$ generated by the two connected components of $\Delta(X)$ do not annihilate each other.

The span $\langle a, b, x \rangle$ is the Matsuo algebra $3C(\frac{1}{2})$ [2, Def. 2.12], so S is obtained from $3C(\frac{1}{2})$ by adjoining an idempotent c that annihilates a and b .

Corollary 1.1. *Each of the following statements, in which the fusion law may be any finite symmetric fusion law, is false. The algebra S of Theorem A is a counterexample to each of (R1)–(R4), and the algebra D of Theorem C is a counterexample to (R5).*

- (R1) *In every primitive axial algebra (A, X) , the subalgebras generated by distinct connected components of $\Delta(X)$ annihilate each other. (This is the form of [10, Conj. 3.16] stated right after it in [10].)*
- (R2) *If a primitive axial algebra (A, X) has only trivial sum decompositions, then $\Delta(X)$ is connected. (This follows from a positive answer to [2, Problem 3.11], [8, Question 9.5].)*
- (R3) *If a primitive axial algebra (A, X) is indecomposable, then $\Delta(X)$ is connected. (The reformulation in [8, §9].)*
- (R4) *If a primitive axial algebra (A, X) is simple, then $\Delta(X)$ is connected.*
- (R5) *In every primitive axial algebra (A, X) , every block I_a , $a \in X$, regarded as an algebra, is indecomposable. ([8, Question 9.2], [2, Problem 3.8].)*

Scope. The law $\mathcal{J}^+(\frac{1}{2})$ is finite and symmetric, but it is not Seress, since $1 \in 0 \star 0$, and it is not $\mathcal{J}(\frac{1}{2})$. So the positive results for Seress fusion laws [7, Thm 6.14], [10, Thm 3.20] and for Jordan type [5, Theorem A] do not apply to it. We claim nothing about the conjecture of [7] for Monster type, about Seress or graded fusion laws, or about simple Majorana algebras [2, Problem 3.12], [8, Question 9.6]; (R4) is only the analogue of that problem for arbitrary finite symmetric fusion laws. The example shows only that (R1)–(R4) need a further hypothesis, on the fusion law or on the algebra; we do not know which hypotheses suffice.

Nested blocks. The *block* I_a of an axis $a \in X$ is the ideal generated by a , and a *dominates* b if $I_b \subseteq I_a$ [8, Defs 3.1, 3.6]. Mamontov, Shpectorov and Zhelyabin ask whether dominance can be non-symmetric [8, Question 9.3], [2, Problem 3.9]; Gorshkov and Shpectorov remark that no example was known, and dominance is symmetric whenever the algebra has a non-degenerate Frobenius form [8, Lemma 6.1]. In Section 4 we point out that a two-dimensional algebra of Peng [11], which also appears in the classification of two-dimensional axial algebras by Kaygorodov, Martín González and Páez-Guillán [6], has $I_b \subsetneq I_a$. Neither paper discusses blocks or dominance; the example is theirs, and we only record that it answers the question. Theorem B in Section 4 gives one more example, for the law $\mathcal{J}^+(\frac{1}{3})$, with a disconnected graph $\Delta(X)$.

A decomposable block. An algebra is *decomposable* if it is the sum of its proper ideals [8, Def. 3.3]. A primitive axial algebra that equals one of its blocks is indecomposable [8, Cor. 3.5], and Mamontov, Shpectorov and Zhelyabin ask whether every block is indecomposable [8, Question 9.2], [2, Problem 3.8], noting that an ideal of a block need not be an ideal of the whole algebra. Theorem C in Section 5 gives, for the finite symmetric law $\mathcal{J}^\circ(\frac{1}{2})$ of Table 1, a five-dimensional primitive axial algebra D whose block I_a is the sum of two proper ideals of I_a , one of which is not an ideal of D . This block is not generated by the axes of X that it contains; the authors of [8] note that a block need not be an axial algebra. If [2, Problem 3.8] is meant only for blocks that are axial algebras in their own right, D does not answer it.

Prior work and literature search. On 27 September 2026, between 17:56 and 18:31 UTC, we searched the arXiv (the version histories of [2, 7, 8, 10, 11], with no newer versions of [2, 8, 10]; metadata; the sources of recent papers on axial algebras); GitHub, including public repositories of AI-assisted work on open problems such as `trureturing` [12]; the problem database MathDB (<https://mathdb.com>), where the conjecture of [7] for Monster type was listed as open; MathOverflow and Mathematics Stack Exchange; OpenAlex and zbMATH Open; home pages of some of the authors and the page of a seminar on axial algebras; and the web. We did not search Google Scholar or MathSciNet, and could not access the published versions of [7, 10], whose arXiv numbering we use. Between 22:59 and 23:23 UTC on the same day we repeated these searches for new material, adding Semantic Scholar’s lists of works citing [2, 8, 10]. On 28 September at 00:30 UTC we checked the arXiv listings and the arXiv API once more (the latest announcement then available was that of 25 September); there was no new paper on axial algebras. We found no earlier example answering [2, Problem 3.11], [8, Question 9.5] or [2, Problem 3.8], [8, Question 9.2] for any fusion law, and for [2, Problem 3.9] only the examples of Section 4. This reports the coverage of our searches, not a guarantee of priority.

2 Definitions

Algebras are commutative and not necessarily associative or unital, over a field $\mathbb{F}$. The statements (R1)–(R5) range over all fields; everything else in this paper, including the lemmas, is over $\mathbb{F} = \mathbb{Q}$, as in the formalization. We write $\langle Y \rangle$ for the linear span and $\langle\langle Y \rangle\rangle$ for the subalgebra generated by a subset Y , and ad_u for the map $v \mapsto uv$.

Definition 2.1. A *fusion law* is a subset $\mathcal{F} \subseteq \mathbb{F}$ with a map $\star: \mathcal{F} \times \mathcal{F} \rightarrow 2^{\mathcal{F}}$ [2, Def. 2.1], [8, Def. 2.1], [10, Def. 2.1]. It is *symmetric* if $\lambda \star \mu = \mu \star \lambda$ for all λ, μ , and *finite* if $\mathcal{F}$ is finite; in [7, Def. 2.1] both are part of the definition. It is *Seress* if $0 \in \mathcal{F}$ and $0 \star \lambda \subseteq \{\lambda\}$ for all $\lambda \in \mathcal{F}$ [10, Def. 3.9].

Definition 2.2. For $a \in A$ and $\lambda \in \mathbb{F}$ let $A_\lambda(a) = \{u \in A : au = \lambda u\}$, and for $\Lambda \subseteq \mathbb{F}$ let $A_\Lambda(a) = \bigoplus_{\lambda \in \Lambda} A_\lambda(a)$, which is 0 for $\Lambda = \emptyset$. A non-zero idempotent a is an $\mathcal{F}$ -*axis* if $A = A_{\mathcal{F}}(a)$ and $A_\lambda(a)A_\mu(a) \subseteq A_{\lambda \star \mu}(a)$ for all $\lambda, \mu \in \mathcal{F}$; it is *primitive* if $A_1(a) = \mathbb{F}a$. A *primitive $\mathcal{F}$ -axial algebra* (A, X) is an algebra A with a set X of primitive $\mathcal{F}$ -axes that generates A [8, Defs 2.2–2.4], [10, Defs 2.2–2.4].

Thus ad_a is semisimple with eigenvalues in $\mathcal{F}$; the set X is part of the data.

Definition 2.3. An *ideal* of an algebra B is a subspace I with $BI \subseteq I$. The algebra B is *decomposable* if it is the sum of its proper ideals [8, Def. 3.3], and *simple* if $B^2 \neq 0$ and 0 and B are its only ideals. For $a \in X$, the *block* I_a is the smallest ideal of A containing a [8, Def. 3.1], [2, Def. 3.7]; dominance is *non-symmetric* if $I_b \subsetneq I_a$ for some $a, b \in X$.

Definition 2.4. The graph $\Delta(X)$ has vertex set X , distinct $a, b \in X$ being adjacent if $ab \neq 0$. A *sum decomposition* of A is a set of subalgebras, any two distinct members $B \neq C$ of which satisfy $BC = 0$, that together generate A ; it is *trivial* if one of its members is A .

$\mathcal{J}(\eta)$	1	0	η	$\mathcal{J}^+(\eta)$	1	0	η	$\mathcal{J}^\circ(\eta)$	1	0	η
1	1		η	1	1		η	1	1		η
0		0	η	0		$1, 0, \eta$	$1, 0, \eta$	0		$0, \eta$	$0, \eta$
η	η	η	$1, 0$	η	η	$1, 0, \eta$	$1, 0$	η	η	$0, \eta$	$1, 0$

Table 1: The Jordan-type law $\mathcal{J}(\eta)$ [2, Fig. 4] and the laws $\mathcal{J}^+(\eta)$ and $\mathcal{J}^\circ(\eta)$; the entry in row λ and column μ lists $\lambda \star \mu$, a blank entry being $\emptyset$. The law $\mathcal{J}^\circ(\eta)$ differs from $\mathcal{J}(\eta)$ in $0 \star 0 = \{0, \eta\}$ and $0 \star \eta = \{0, \eta\}$.

The law $\mathcal{J}^+(\frac{1}{2})$ is finite and symmetric, and it is not Seress, since $1 \in 0 \star 0$; the same holds for $\mathcal{J}^+(\frac{1}{3})$; the law $\mathcal{J}^\circ(\frac{1}{2})$ is finite and symmetric and not Seress, since $\frac{1}{2} \in 0 \star 0$. The next lemma shows that for $\mathcal{J}^+(\eta)$ only the rule $\eta \star \eta$ needs checking.

Lemma 2.5. *Let $\eta \neq 0, 1$, and let a be a non-zero idempotent of an algebra A with $A = A_1(a) \oplus A_0(a) \oplus A_\eta(a)$ and $A_1(a) = \mathbb{F}a$. If $A_\eta(a)A_\eta(a) \subseteq A_1(a) \oplus A_0(a)$, then a is a primitive $\mathcal{J}^+(\eta)$ -axis.*

Proof. Since $A_1(a) = \mathbb{F}a$, we have $A_1(a)A_\lambda(a) = \lambda A_\lambda(a)$, which lies in $A_{1 \star \lambda}(a)$ for $\lambda = 1, 0, \eta$ (for $\lambda = 0$ it is $0 = A_0(a)$). The entries $0 \star 0$ and $0 \star \eta$ are all of $\{1, 0, \eta\}$, so they impose no condition, and $\eta \star \eta = \{1, 0\}$ is the hypothesis. $\square$

3 The algebra S : proof of Theorem A

(a) **Axes and generation.** Table 2 lists eigenvectors of ad_a , ad_b and ad_c , where

$$w = 2(x - a - b) + c.$$

For each of the three axes the four vectors listed form a basis of S , so $S = A_1 \oplus A_0 \oplus A_{1/2}$ with the eigenspaces spanned by the listed vectors; in particular each axis is primitive. For instance, $a(b + x - \frac{1}{2}a) = \frac{1}{4}(a + b - x) + \frac{1}{4}(a - b + x) - \frac{1}{2}a = 0$, $a(x - b) = \frac{1}{2}(x - b)$ and $cw = 2cx + c = (x - a - b) + \frac{1}{2}c = \frac{1}{2}w$. The last column gives the product of the $\frac{1}{2}$ -eigenvectors, which lies in $A_1 \oplus A_0$. The row of b is checked in the same way as that of a . By Lemma 2.5, a , b and c are primitive $\mathcal{J}^+(\frac{1}{2})$ -axes. Finally $x = a + b - 4ab \in \langle\langle a, b \rangle\rangle$, so X generates S .

axis	A_1	A_0	$A_{1/2}$	square of the $\frac{1}{2}$ -eigenvector
a	a	$b + x - \frac{1}{2}a, c$	$x - b$	$(x - b)^2 = \frac{3}{4}a + \frac{1}{2}(b + x - \frac{1}{2}a)$
b	b	$a + x - \frac{1}{2}b, c$	$x - a$	$(x - a)^2 = \frac{3}{4}b + \frac{1}{2}(a + x - \frac{1}{2}b)$
c	c	a, b	w	$w^2 = 4a + 4b$

Table 2: Eigenvectors of the three axes of S , with $w = 2(x - a - b) + c$.

(b) Simplicity. Let $P_c = 2\text{ad}_c^2 - \text{ad}_c$. It fixes c and kills $A_0(c)$ and $A_{1/2}(c)$. Writing $u = \alpha a + \beta b + \xi x + \gamma c$ as $u = (\alpha + \xi)a + (\beta + \xi)b + \frac{\xi}{2}w + (\gamma - \frac{\xi}{2})c$, we get

$$P_c(u) = \varphi(u)c, \quad \varphi(u) = \gamma - \frac{1}{2}\xi.$$

The four linear forms $u \mapsto \varphi(u), \varphi(au), \varphi(bu), \varphi(xu)$ have coefficient vectors, with respect to $(\alpha, \beta, \xi, \gamma)$,

$$(0, 0, -\frac{1}{2}, 1), \quad (0, \frac{1}{8}, -\frac{1}{8}, 0), \quad (\frac{1}{8}, 0, -\frac{1}{8}, 0), \quad (-\frac{1}{8}, -\frac{1}{8}, -\frac{1}{2}, -\frac{1}{2}),$$

and they vanish simultaneously only at $u = 0$: the second and third give $\beta = \xi$ and $\alpha = \xi$, the first gives $\gamma = \xi/2$, and then the fourth equals $-\xi$. Let $I \neq 0$ be an ideal and $0 \neq u \in I$. Then one of the elements $v = u, au, bu, xu$ of I has $\varphi(v) \neq 0$, and $P_c(v) = 2c(cv) - cv \in I$ gives $c \in I$. Next, $xc = -\frac{1}{2}a - \frac{1}{2}b + \frac{1}{2}x - \frac{1}{4}c$, $a(xc) = -\frac{1}{2}a - \frac{1}{4}b + \frac{1}{4}x$ and $b(xc) = -\frac{1}{4}a - \frac{1}{2}b + \frac{1}{4}x$ lie in I , and

$$a = \frac{1}{2}c + 2xc - 4a(xc), \quad b = \frac{1}{2}c + 2xc - 4b(xc), \quad x = a + b - 4ab.$$

So $I = S$. Since $c^2 = c \neq 0$, we have $S^2 \neq 0$, and S is simple.

(c) The graph. We have $ab = \frac{1}{4}(a + b - x) \neq 0$ and $ac = bc = 0$. So $a-b$ is the only edge of $\Delta(X)$, whose connected components are $\{a, b\}$ and $\{c\}$.

(d) Sum decompositions. This follows from (b) and the next lemma.

Lemma 3.1. *Every sum decomposition of a simple algebra A is trivial.*

Proof. Let $\mathcal{S}$ be a sum decomposition and $W = \sum_{B \in \mathcal{S}} B$, the sum as subspaces. A product of an element of B and an element of C lies in B if $B = C$ and is 0 otherwise, so W is a subalgebra; it contains every member of $\mathcal{S}$, hence $W = A$. For $B \in \mathcal{S}$ and $v \in B$, write $u \in A$ as $u = \sum_C u_C$ with $u_C \in C$; then $uv = u_B v \in B$. So each member of $\mathcal{S}$ is an ideal, hence 0 or A . If none were A , then $A = W = 0$, contradicting $A^2 \neq 0$. $\square$

(e) The components do not annihilate each other. We have $x \in \langle\langle a, b \rangle\rangle$ and $c \in \langle\langle c \rangle\rangle$, but $xc = \frac{1}{2}(x - a - b) - \frac{1}{4}c \neq 0$. This completes the proof of Theorem A. $\square$

Proof of Corollary 1.1. We need one more simple fact.

Lemma 3.2. *A simple algebra is indecomposable.*

Proof. Its only proper ideal is 0, and $A \neq 0$ since $A^2 \neq 0$. $\square$

The law $\mathcal{J}^+(\frac{1}{2})$ is finite and symmetric, and by Theorem A the pair (S, X) is a primitive $\mathcal{J}^+(\frac{1}{2})$ -axial algebra whose graph $\Delta(X)$ is not connected. Moreover, by Theorem A, the subalgebras generated by the two components of $\Delta(X)$ do not annihilate each other, which refutes (R1); S has only trivial sum decompositions, which refutes (R2); and S is simple, which refutes (R4), and hence, by Lemma 3.2, indecomposable, which refutes (R3). Statement (R5) is refuted by Theorem C in Section 5, since $\mathcal{J}^\circ(\frac{1}{2})$ is finite and symmetric. $\square$

4 Nested blocks

Let P be the two-dimensional algebra over $\mathbb{Q}$ with basis a, b and products $a^2 = a$, $ab = 2b$, $b^2 = b$, and let $X = \{a, b\}$. Let $\mathcal{F}_{D3}$ be the fusion law on $\{0, 1, 2\}$ with

$$0 \star 0 = \{0, 1\}, \quad 1 \star 1 = \{1\}, \quad 1 \star 2 = 2 \star 1 = 2 \star 2 = \{2\},$$

all other entries being empty. This algebra and fusion law are those of Peng [11], who shows that (P, X) is a primitive $\mathcal{F}_{D3}$ -axial algebra whose only ideals are 0 , $\langle b \rangle$ and P [11, Thm 2.1], and identifies (P, X) with an axial algebra in the classification of two-dimensional axial algebras of [6] [11, Prop. 3.1]. The following is therefore immediate from [11]; we include the short proof.

Proposition 4.1. *(P, X) is a primitive $\mathcal{F}_{D3}$ -axial algebra with $I_b = \langle b \rangle \subsetneq I_a = P$. In particular, dominance in (P, X) is non-symmetric.*

Proof. The map ad_a has eigenvectors a (eigenvalue 1) and b (eigenvalue 2), and ad_b has eigenvectors b (eigenvalue 1) and $a - 2b$ (eigenvalue 0). The products of eigenvectors are $b^2 = b \in A_2(a)$, $b(a - 2b) = 0$ and $(a - 2b)^2 = a - 4b = (a - 2b) - 2b \in A_{\{0,1\}}(b)$, as the law requires, so a and b are primitive $\mathcal{F}_{D3}$ -axes, and they span P . The subspace $\langle b \rangle$ is an ideal, and an ideal containing a contains $ab = 2b$. $\square$

The law $\mathcal{F}_{D3}$ is finite and symmetric. So Proposition 4.1 gives a positive answer to [8, Question 9.3], [2, Problem 3.9], with a finite symmetric fusion law. We do not claim it as new: the algebra is in [6, 11], which do not discuss blocks or dominance.

Theorem B. *Let E be the four-dimensional commutative algebra over $\mathbb{Q}$ with basis a, b, x, c , in which a, b, x span the Matsuo algebra $3C(\frac{1}{3})$: $a^2 = a$, $b^2 = b$, $x^2 = x$, $ab = \frac{1}{6}(a + b - x)$, $ax = \frac{1}{6}(a - b + x)$, $bx = \frac{1}{6}(-a + b + x)$, and $c^2 = c$, $ac = bc = 0$, $cx = \frac{1}{3}(x - a - b)$. Let $X = \{a, b, c\}$. Then (E, X) is a primitive $\mathcal{J}^+(\frac{1}{3})$ -axial algebra with $I_a = I_b \subsetneq I_c = E$; the algebra E is indecomposable, $\Delta(X)$ is not connected, and the subalgebras generated by its two connected components do not annihilate each other.*

Proof. For a we have $A_1(a) = \langle a \rangle$, $A_0(a) = \langle b + x - \frac{1}{3}a, c \rangle$ and $A_{1/3}(a) = \langle x - b \rangle$, with $(x - b)^2 = \frac{2}{3}(b + x - \frac{1}{3}a) + \frac{5}{9}a$; for b we have $A_1(b) = \langle b \rangle$, $A_0(b) = \langle a + x - \frac{1}{3}b, c \rangle$ and $A_{1/3}(b) = \langle x - a \rangle$, with $(x - a)^2 = \frac{2}{3}(a + x - \frac{1}{3}b) + \frac{5}{9}b$. For c we have $A_1(c) = \langle c \rangle$, $A_0(c) = \langle a, b \rangle$ and $A_{1/3}(c) = \langle v \rangle$ with $v = x - a - b$, $cv = \frac{1}{3}v$ and $v^2 = \frac{4}{3}(a + b)$. By Lemma 2.5, a, b, c are primitive $\mathcal{J}^+(\frac{1}{3})$ -axes, and $x = a + b - 6ab$, so X generates E .

The subspace $W = \langle a, b, x \rangle$ is an ideal, since $cW \subseteq \langle v \rangle$, so $I_a, I_b \subseteq W \neq E$. Since $b = 9b(ab) - 3ab$ and $a = 9a(ab) - 3ab$, each of I_a and I_b contains a and b , so $I_a = I_b$. An ideal containing c contains $v = 3cx$, then $xv = \frac{2}{3}x$, so x , then $a + b = x - v$ and $a - b = 6ax - x$; so $I_c = E$, and $I_a \subsetneq I_c$. The map $\frac{1}{2}(3\text{ad}_c^2 - \text{ad}_c)$ sends $u = \alpha a + \beta b + \xi x + \gamma c$ to γc , so an ideal containing an element with $\gamma \neq 0$ contains c and is E . Hence every proper ideal lies in W , and E is not the sum of its proper ideals. Finally $ab \neq 0$, $ac = bc = 0$, $x \in \langle\langle a, b \rangle\rangle$ and $xc = \frac{1}{3}(x - a - b) \neq 0$. $\square$

Since the law $\mathcal{J}^+(\frac{1}{3})$ is finite and symmetric, E is a further example of non-symmetric dominance for such laws, and also a counterexample to (R1) and (R3) of Corollary 1.1.

5 A decomposable block

Theorem C. *Let D be the five-dimensional commutative algebra over $\mathbb{Q}$ with basis a, b, x, c, d in which a, b, x span the Matsuo algebra $3C(\frac{1}{2})$, with the products of Theorem A, and*

$$c^2 = c, \quad ac = bc = 0, \quad cx = d, \quad cd = \frac{1}{2}d, \quad ad = bd = xd = d^2 = 0.$$

Let $X = \{a, b, c\}$. Then:

- (a) (D, X) is a primitive $\mathcal{J}^\circ(\frac{1}{2})$ -axial algebra;
- (b) the block I_a is $\langle a, b, x, d \rangle$;
- (c) $\langle a, b, x \rangle$ and $\langle d \rangle$ are ideals of the algebra I_a , and I_a is decomposable;
- (d) $\langle a, b, x \rangle$ is not an ideal of D ;
- (e) D has a Frobenius form, that is, a bilinear form β with $\beta(uv, w) = \beta(u, vw)$ for all u, v, w , such that $\beta(a, a)$, $\beta(b, b)$ and $\beta(c, c)$ are non-zero.

Proof. (a) For a we have $A_1(a) = \langle a \rangle$, $A_0(a) = \langle b + x - \frac{1}{2}a, c, d \rangle$ and $A_{1/2}(a) = \langle x - b \rangle$. On $\langle a, b, x \rangle$ the products of these eigenvectors obey $\mathcal{J}^\circ(\frac{1}{2})$, as in any Matsuo algebra [2, §2.1]; the new products are $(b + x - \frac{1}{2}a)c = (x - b)c = d \in A_0(a)$, $c^2 = c$ and $cd = \frac{1}{2}d$, and d annihilates a, b, x, d . So $A_0A_0 \subseteq A_0$, $A_0A_{1/2} \subseteq A_0 \oplus A_{1/2}$ and $A_{1/2}A_{1/2} \subseteq A_1 \oplus A_0$ for the axis a , as $\mathcal{J}^\circ(\frac{1}{2})$ requires. For b we have $A_1(b) = \langle b \rangle$, $A_0(b) = \langle a + x - \frac{1}{2}b, c, d \rangle$ and $A_{1/2}(b) = \langle x - a \rangle$, and the same argument applies, with $(a + x - \frac{1}{2}b)c = (x - a)c = d \in A_0(b)$. For c we have $A_1(c) = \langle c \rangle$, $A_0(c) = \langle a, b, x - 2d \rangle$ and $A_{1/2}(c) = \langle d \rangle$, with

$$\begin{aligned} ab &= \frac{1}{4}(a + b - (x - 2d)) - \frac{1}{2}d, & a(x - 2d) &= \frac{1}{4}(a - b + (x - 2d)) + \frac{1}{2}d, \\ b(x - 2d) &= \frac{1}{4}(-a + b + (x - 2d)) + \frac{1}{2}d, & (x - 2d)^2 &= (x - 2d) + 2d, \end{aligned}$$

and $a^2 = a$, $b^2 = b$; all these lie in $A_0(c) \oplus A_{1/2}(c)$, while $a, b, x - 2d$ and d annihilate d . So a, b, c are primitive $\mathcal{J}^\circ(\frac{1}{2})$ -axes, and $x = a + b - 4ab$, $d = cx$ show that X generates D .

(b) The subspace $\langle a, b, x, d \rangle$ is an ideal. An ideal containing a contains $ab - ax = \frac{1}{2}(b - x)$, then $b(b - x) - \frac{1}{4}a - \frac{1}{4}(b - x) = \frac{1}{2}b$, so b and x , and then $d = cx$.

(c) The subspace $\langle a, b, x \rangle$ is a subalgebra annihilated by d , and $\langle d \rangle$ is annihilated by a, b, x, d ; so both are ideals of I_a . They are proper, and their sum is I_a .

(d) $cx = d \notin \langle a, b, x \rangle$.

(e) Let β be the symmetric bilinear form with $\beta(a, a) = \beta(b, b) = \beta(x, x) = \beta(c, c) = 1$, $\beta(a, b) = \beta(a, x) = \beta(b, x) = \frac{1}{4}$ and $\beta(y, z) = 0$ for all other pairs of basis vectors. On $\langle a, b, x \rangle$ it is the Frobenius form of the Matsuo algebra [2, §2.1], and $\beta(uv, w) = \beta(u, vw)$ holds for all 125 triples of basis vectors. $\square$

Remark 5.1. (1) For primitive axial algebras with a non-degenerate Frobenius form, every block is a simple algebra [8, Prop. 6.2]. Statement (e) requires only that the three generating axes have non-zero self-pairings. (2) The axes of X in I_a are a and b , since the c -coordinate of every element of I_a is 0, and they generate only $\langle a, b, x \rangle \subsetneq I_a$. So the block I_a is not generated by the axes of X that it contains; compare the remark after [8, Cor. 3.5]. (3) The law $\mathcal{J}^\circ(\frac{1}{2})$ is finite and symmetric but not Seress, since $\frac{1}{2} \in 0 \star 0$.

6 Formal verification

The results are formally verified in Lean 4 (version 4.34.1) with Mathlib [9] (version v4.34.1, commit d13f23b7).

Statements before proofs. After the examples had been found and checked by exact computation, the agent that found them wrote a Lean file of definitions and statements, the *statement file*. It contains no proofs, apart from the bilinearity of products given by structure constants. An independent agent reviewed version 1 of this file against the source papers and recomputed the examples; its corrections were applied in version 2, which in particular restricts the general statements to finite symmetric fusion laws. After a second independent review, version 2 was frozen on 27 September 2026 at 21:42 UTC by recording its SHA-256 hash 0d88e7321e0f0931f953ea250f2c5dc1d6d5773cd8f78e3080e0d6eeceb13ee1. Known-answer

tests of the definitions were written with version 2. The proofs listed below were written only after the freeze, by a different agent, in separate files that import the statement file, which was not changed. Two further statements about the algebra D , used in Remark 5.1(2), were added later in a separate file: the agent that wrote the proofs wrote them, an independent agent of the same system reviewed them, and they were frozen on 27 September 2026 at 23:55 UTC (SHA-256 `afd751c0df8ba036995314a0efd7a5b60d6b333ab9958111a879724d81fc9175`) before their proofs were written.

What is proved. Table 3 lists the proved declarations. In ASCII notation (`forall`, `->`, `/\`, `Not` and `Rat` for $\forall$, $\rightarrow$, $\wedge$, $\neg$ and $\mathbb{Q}$, and `->1[K]` for K -linear maps), the general form of (R1) and the statement of Theorem A read as follows.

```
def MS_Conjecture_3_16 : Prop :=
  forall (K : Type) [Field K] (V : Type) [AddCommGroup V] [Module K V]
    (F : FusionLaw K) (mu : V ->1[K] V ->1[K] V) (X : Set V),
    F.IsFiniteSymmetric -> IsPrimitiveAxialAlgebra F mu X ->
      ComponentsAnnihilate mu X

def ExS.Statement : Prop :=
  IsPrimitiveAxialAlgebra (JPlus (1/2 : Rat)) mu X /\ IsSimpleAlg mu /\
    Not (nonAnnGraph mu X).Connected /\ OnlyTrivialSumDecompositions mu /\
    Not (ComponentsAnnihilate mu X)
```

In the second definition, μ is the product of S on $\mathbb{Q}^4$, given by structure constants in the basis $e_0, e_1, e_2, e_3 = a, b, x, c$, and X is $\{e_0, e_1, e_3\}$. The general forms of (R2), (R3) and (R4), namely `FinestSumDecomposition_connected`, `Indecomposable_connected` and `Simple_connected`, have the same binders and the same two hypotheses as `MS_Conjecture_3_16`, then the hypothesis `OnlyTrivialSumDecompositions mu`, `Not (IsDecomposable mu)` or `IsSimpleAlg mu`, respectively, and the conclusion `(nonAnnGraph mu X).Connected`. The general form of (R5), `Blocks_indecomposable`, has the same binders and two hypotheses and the conclusion that `IsDecomposableSub mu (block mu a)` fails for every $a \in X$; the remaining part of `Corollary_Dominance_nonsymmetric`, states that for some field, space, finite symmetric fusion law and primitive axial algebra (V, X) there are $a, b \in X$ with `block mu a < block mu b`.

Checks. For the declarations of Table 3, the recorded axiom audits (`#print axioms`) report only the axioms `propext`, `Classical.choice` and `Quot.sound`. A static scan found no `sorry`, `admit`, `native_decide`, `bv_decide`, `implemented_by`, `extern`, `unsafe` or `axiom` in the proof files. The Lean package of this paper (the statement files, the known-answer tests, 20 proof modules, the file of Table 3 and an audit file) was then built from an empty directory, reusing only the pinned toolchain and the compiled Mathlib and its dependencies, one module at a time with warnings treated as errors, and every module was checked with `leanchecker`, all with exit code 0 (28 September 2026, between 00:00 and 00:10 UTC). For the release, the whole Lean project of the repository, which also contains the formalizations for earlier papers, was rebuilt from a clean directory (28 September 2026, 00:25–01:05 UTC, exit code 0), and all its modules, including those of this paper, were checked again with `leanchecker`; every axiom list printed in that rebuild is a subset of the three axioms above. The Lean files are in version 1.8.0 of the repository [1], directory `lean/Research/AxialMSZ/`.

Definitions the reader must trust. They are listed in Table 4; compare Section 2.

Faithfulness and non-claims. An algebra is modelled as a vector space V over a field K with a bilinear map $V \rightarrow V \rightarrow V$; commutativity is a separate hypothesis, and associativity and a unit are never assumed. A fusion law is a subset of K with a set-valued map `star` on $K \times K$; the fusion rules of an axis use only its values on the carrier. A finite symmetric

Lean theorem	its type, and what it says	here
check_Laws_facts	Laws_facts : $\mathcal{J}^+(\frac{1}{2})$, $\mathcal{J}^+(\frac{1}{3})$, $\mathcal{J}^\circ(\frac{1}{2})$, $\mathcal{F}_{D3}$ are finite and symmetric (with values in the carrier); the first three are not Seress	§§2, 4
check_TheoremA	TheoremA , which is ExS.Statement above	Thm A
check_not_MS_Conjecture_3_16	$\neg$ MS_Conjecture_3_16 : (R1) is false	Cor. 1.1
check_not_FinestSumDecomposition_connected	$\neg$ FinestSumDecomposition_connected : (R2) is false	Cor. 1.1
check_not_Indecomposable_connected	$\neg$ Indecomposable_connected : (R3) is false	Cor. 1.1
check_not_Simple_connected	$\neg$ Simple_connected : (R4) is false	Cor. 1.1
check_RemarkP	RemarkP , which is ExP.Statement : (P, X) is a primitive $\mathcal{F}_{D3}$ -axial algebra and $I_b \subsetneq I_a$	Prop. 4.1
check_Dominance_nonsymmetric	Dominance_nonsymmetric : some primitive axial algebra with a finite symmetric fusion law has axes a, b with $I_a \subsetneq I_b$	§4
check_TheoremB	TheoremB , which is ExE.Statement : Theorem B, with the blocks stated as $I_a \subsetneq I_c$, $I_a = I_b$ and $I_c = E$	Thm B
check_TheoremC	TheoremC , which is ExD.Statement : Theorem C (a)–(e)	Thm C
check_ExD_axes_in_block [†]	the axes of X in I_a are a and b	Rem. 5.1
check_ExD_block_not_axial_on_contained_axes [†]	a and b generate $\langle a, b, x \rangle$, and $\langle a, b, x \rangle \subsetneq I_a$	Rem. 5.1
check_Corollary	Corollary : (R1)–(R5) are false, and for a finite symmetric fusion law some primitive axial algebra has non-symmetric dominance	Cor. 1.1, §4

Table 3: The proved declarations, in the namespace **AxialMSZ.Check**; each proves a declaration of the frozen statement file (namespace **AxialMSZ.Challenge**) or its negation, or, if marked [†], of the supplementary file (namespace **CodexAxial.SupplementChallenge**).

Lean definition	meaning
FusionLaw K	a carrier $\mathcal{F} \subseteq K$ and a map star : $K \rightarrow K \rightarrow \text{Set } K$; the carrier is finite, star is symmetric, and its values on the carrier lie in the carrier; $0 \in \mathcal{F}$
IsFiniteSymmetric	and $0 * \lambda \subseteq \{\lambda\}$ for $\lambda \in \mathcal{F}$
IsSeress	the laws $\mathcal{J}^+(\eta)$, $\mathcal{F}_{D3}$, $\mathcal{J}^\circ(\eta)$ as if-then-else tables
JPlus , FD3 , JMild	Mathlib’s eigenspace $\ker(\text{ad}_a - \lambda)$, and the supremum of these over $\lambda \in \Lambda$ (0 if $\Lambda = \emptyset$)
eigsp , eigspSet	
IsAxis	$a \neq 0$, $a^2 = a$, $A_{\mathcal{F}}(a) = A$ and $A_\lambda(a)A_\mu(a) \subseteq A_{\lambda * \mu}(a)$ for $\lambda, \mu \in \mathcal{F}$; in addition $A_1(a) = \langle a \rangle$
IsPrimitiveAxis	the smallest subspace containing X and closed under the product; the product is commutative, every $a \in X$ is a primitive $\mathcal{F}$ -axis, and X generates
gen	
IsPrimitiveAxialAlgebra	
IsIdeal , block	a subspace I with $uv, vu \in I$ for all u and all $v \in I$; the smallest ideal containing a ; the sum of all ideals other than A is A ; some product is non-zero and every ideal is 0 or A
IsDecomposable	
IsSimpleAlg	
IsIdealIn	an ideal of a subalgebra B regarded as an algebra; B is the sum of its ideals other than B ; a bilinear form with $\beta(uv, w) = \beta(u, vw)$ for all u, v, w
IsDecomposableSub	
IsFrobeniusForm	
nonAnnGraph	Mathlib’s SimpleGraph.fromRel of $uv \neq 0$ on the subtype X , with Mathlib’s connectedness and components
IsSumDecomposition	a set of subalgebras, distinct members multiplying to 0, whose union generates A ; every such set contains A ; $uv = 0$ whenever u and v lie in the subalgebras generated by two distinct connected components
OnlyTrivialSumDecompositions	
ComponentsAnnihilate	
structProduct , ExS.T , ExP.T , ExE.T , ExD.T	the product on $\mathbb{Q}^n$ given by a table of structure constants; the tables of S , E , D and P

Table 4: The definitions of the statement file used in Table 3.

law in Lean must be symmetric on all of $K \times K$, with its values on the carrier inside the carrier; restricted to the carrier it is finite and symmetric in the sense of Definition 2.1, so the

negations proved in Lean give the negations of the statements of Corollary 1.1 as stated. The general statements quantify over all fields and vector spaces in the lowest universe and over all finite symmetric fusion laws, as in [7, Def. 2.1]; so their negations imply the corresponding negative statements for the arbitrary fusion laws of [2, 8] and the symmetric ones of [10]. The graph $\Delta(X)$ is built with `SimpleGraph.fromRel`, which symmetrises the relation $uv \neq 0$; for a commutative product this changes nothing. Sum decompositions are sets of subalgebras rather than indexed families; “every sum decomposition contains A ” implies the same statement for families. For Theorem C, the ideals of a block are those of the block regarded as an algebra, which need not be ideals of D , as in the remark after [8, Cor. 3.5]; a Frobenius form is not required to be symmetric or non-degenerate. The tables of structure constants were compared with the displayed products by the reviewing agents, and again by us with a separate script. The formalisation says nothing about Seress, graded or Monster-type fusion laws or about Majorana algebras, and `Simple_connected` is the analogue of [2, Problem 3.12] for finite symmetric fusion laws, not that problem. That the Lean definitions agree with those of the sources was checked by the reviewing agents, not by a machine.

How this paper was produced

This work was carried out on 27 and 28 September 2026 (UTC) in a workflow directed by the author, using AI agents. Agents of Anthropic Claude Code, with the model Claude Opus 5.5, selected the questions, found the examples and their proofs, checked the examples in exact rational arithmetic with two separately written programs, wrote the statement file of Section 6, carried out the literature search described in the introduction, and drafted this text. An independent Claude Code agent, which had not taken part in this work, reviewed version 1 of the statement file against the source papers and recomputed the examples with its own program; its corrections were applied in version 2. OpenAI Codex then reviewed the mathematics and version 2 of the statement file independently, recomputing the examples with its own program, wrote the Lean proofs and ran the checks described in Section 6; it also wrote the two supplementary statements of Section 6, which an independent Codex agent reviewed before they were frozen. Codex also reviewed a draft of this paper, recomputing its computations with its own program, and its corrections were applied. Throughout, “we” refers to this workflow. The author is not a professional mathematician. The author has read the paper and, using a side-by-side table, compared the statements of the Lean theorems with the results stated here. No human mathematician has reviewed this paper. The correctness of the results rests on the Lean proofs of the statements in Table 3 and on the agreement of those statements with the definitions of the source papers, which was checked by the reviewing agents; the written proofs are given for the reader. AI systems are not authors of this paper; the author takes full responsibility for its content.

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